AI Engineer — Technical Assessment

REJ Investment + Cherry Host Platform

Thanks for taking the time to interview with us. Before we move forward, we'd love to see a small sample of your thinking and your work. This assessment should take approximately **3–4 hours total**. We value quality thinking over speed — please take your time and submit within **72 hours**.

The Platform – Two Brands, One System

We've built a complete real estate and property management platform that runs **two brands** from one codebase, one React + TypeScript app, and one Supabase database with **228 tables**.

🏢 REJ Investment

my-investments.rej.co.il — A platform for Israeli investors buying properties in Madrid. Features a 7-page PDF investment calculator with ROI/IRR analysis, a deal pipeline from sourcing to acquisition, and an investor portal with portfolio views, shared simulations, and documents. Marketing site is in Hebrew.

🍒 Cherry Host

cherry-host.es — A student rental platform in Madrid. Students search properties, view details, and submit applications (max 3 per property, queue system). Backend includes lease management, rent tracking, a room-by-room check-in/out system, maintenance ticketing, and a full tenant portal.

ⓘ Both brands share one admin panel — a FeedHive-style CMS with a dark navy collapsible sidebar. Domain detection drives branding: `cherry-host` in hostname → Cherry Host UI; otherwise → REJ Hebrew branding. Stack: **React + TypeScript + Tailwind · Supabase (PostgreSQL, Edge Functions, Auth, Realtime, Storage) · Resend · Twilio**

The 8 AI Agents - Your Playground

The admin panel is organized around **8 department "agents"**, each with its own dashboards, pages, and data. Every agent also shares a common toolkit: WhatsApp-style messaging, Monday.com-style Kanban tasks, Google-synced calendar, email campaigns, voice notes, AI chat, and Twilio calls.

🏗️ Mila — Acquisition & Investment

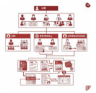
Scans Idealista, Fotocasa, and auction sites. Each property is analyzed with **80+ parameters** and an AI score. Supports Fix&Flip, Rent-to-Rent, and Student Rental models. Manages the full owned portfolio and blog.

🏷️ Daniel — Property Sales

A **9-stage Kanban pipeline** from Preparation to Notary Completed. Includes a viewings calendar, offers comparison board, and auto-publishing to Idealista, Fotocasa, and Facebook.

💰 Noah — Finance & Billing

Full accounting system with multi-company support. Bank import with match/classify/post actions, expense management with receipt upload, IVA/IRPF invoice creation, and **30+ financial reports** including Modelo 303/111/200.



👥 Eva — HR

Employee directory, org chart, global clock-in/out, attendance tracking, vacation calendar, recruitment Kanban, employee evaluations, payslip sending, and document e-signatures — like Sesame HR built in.

📢 Maria — Marketing

AI Video Studio, Image Studio (15+ styles), Voice Studio (ElevenLabs), and Content Writer. Multi-platform publishing to Instagram, Facebook, LinkedIn, TikTok, YouTube, and Twitter. Brand Kit per company.

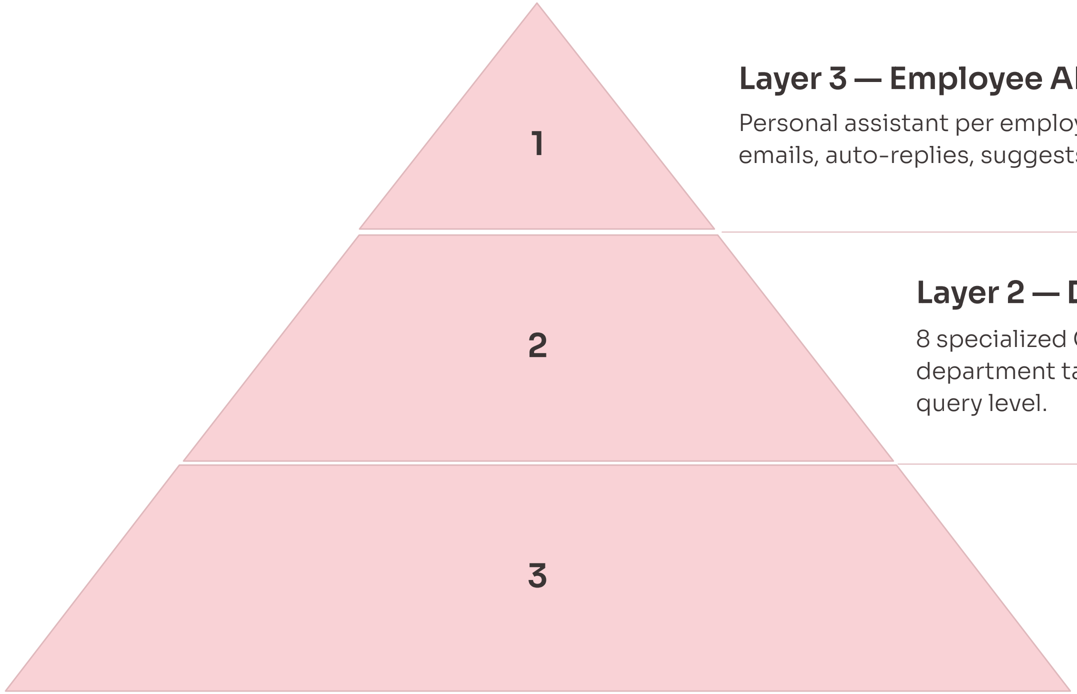
🏠 Sofia — Rental Operations

Vacancy tracker, application queue, lease contracts, rent tracking, maintenance ticketing, and the room-by-room check-in/out system where missing items are auto-deducted from tenant deposits.

🔜 **Alex — Investment Leads** and **Milo — Rent-to-Rent** are next on the roadmap and will be built as part of the platform expansion.

The AI Architecture - What You'll Build

You'll design and implement **three interconnected layers of AI** that power the entire platform — from a central orchestrating brain down to every individual employee's personal assistant.



Layer 3 — Employee AI

Personal assistant per employee using pgvector embeddings. Learns communication style after 500+ messages: drafts emails, auto-replies, suggests responses in the employee's own voice.

Layer 2 — Department Agent Brains

8 specialized Claude-powered AIs, each with a unique system prompt and access ONLY to its own department tables. Noah never sees HR data. Eva never sees financials. Strict isolation is enforced at the query level.

Layer 1 — Platform Brain (Central AI)

One Claude API brain with visibility across ALL companies, ALL properties, ALL departments. Orchestrates cross-department analysis, generates company-wide reports, and runs **AI Agent Meetings** — a virtual strategy session where all 8 agents discuss real numbers.

🤖 AI Auto-fill

Enter address + photos → AI generates full property description, nearby metro, universities, amenities, per-room inventory with replacement costs.

💬 Chat Widget

Multilingual AI chat on cherry-host.es for website visitors — English, Spanish, and Hebrew supported out of the box.

📋 Meeting Summaries

After video calls, AI auto-transcribes and extracts action items, owners, and deadlines — no manual notes needed.

🔍 Competitor Monitor

Auto-monitors competitor websites and social channels via OpenClaw browser automation. AI generates strategic recommendations from the data.

Questions + Practical Task

WRITTEN ASSESSMENT + CODE SUBMISSION

1

Agent Isolation

Each of the 8 agents needs access to **different database tables**. Noah sees finances, Eva sees HR, Sofia sees rentals — they should never cross. How would you architect this so each agent only accesses its own data? Consider **system prompts, function calling, and security**. Answer in 5–10 sentences.

2

AI Agent Meeting

All 8 agents sit in a virtual meeting and discuss strategy with **real data**: Noah says "Revenue is up 12%", Sofia says "3 vacancies this month", Maria says "Instagram dropped 30%." How do you **orchestrate this conversation**? How do you prevent hallucinated numbers? Answer in 5–10 sentences.

3

Scale & Rate Limits

15 employees use personal AI assistants simultaneously, each sending 2–3 requests per minute. Claude API has rate limits. How do you handle this? Think about **queuing, caching, and cost optimization**. Answer in 5–10 sentences.

🛠️ **Practical Task (2–3 hours):** Build a TypeScript application with two AI agents — **"Finance Agent"** and **"HR Agent"** — each with a unique system prompt and hardcoded mock JSON data (no DB needed). Build a routing function that receives a user question and sends it to the correct agent. **Bonus:** User asks "Should we hire more people?" → both agents discuss and produce a joint recommendation. Use Claude or OpenAI API. Submit code + README. We evaluate clean TypeScript, smart prompts, routing logic, and README quality. No UI or production code expected.

Ready to Submit? 🚀

We're excited to see how you think — both your written reasoning and your hands-on code tell us a lot about how you'd approach building on this platform. Good luck!

📝 Written Answers

Thoughtful responses to all 3 questions, 5–10 sentences each. Focus on your reasoning and the trade-offs you'd consider — not just the happy path.

💻 Code + README

TypeScript application with Finance Agent and HR Agent, a routing function, and a clear README explaining your design decisions and how to run the project.

⏰ Deadline

Submit within **72 hours** of receiving this assessment. Quality over speed — take the time you need to produce work you're proud of.

> We value how you think, architect, and communicate — as much as the code itself. Show us your best work.